

Experiment 5: Advanced - Pipelined RISC-V Processor

1. Objective

The objective of this experiment is to design, implement, and verify a 32-bit pipelined RISC-V processor supporting the RV32I Instruction Set Architecture (ISA). This advanced design aims to improve instruction throughput compared to a single-cycle processor by executing multiple instructions concurrently across different pipeline stages, while effectively handling data and control hazards.

2. Block Diagram

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mermaid version 12.1.0

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'SHAPE_DATA', 'STYLE_SEPARATOR', 'START_LINK', 'LINK', 'LINK_ID', got
TAGSTART